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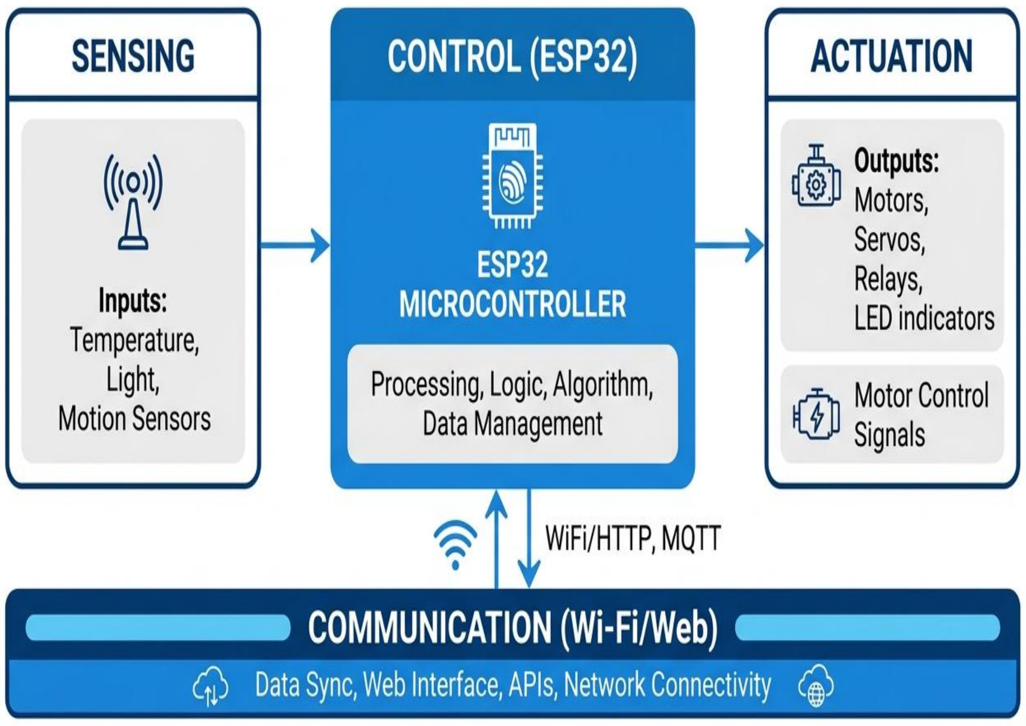
Smart Multi-Mode IoT-Based Robotic Vehicle

ESP32-enabled platform for manual control, obstacle avoidance, and smart exploration

Introduction

IoT robotics blends sensing, connectivity, and autonomy for real-world tasks
ESP32 enables low-cost wireless control with sufficient compute for edge decisions
Multi-mode vehicles reduce mission risk by switching behaviors on demand
Applications: indoor logistics, labs, inspection, education, and search scenarios



Problem Statement

Existing low-cost robots often lack reliable mode switching and remote observability

Manual-only control is unsafe in cluttered environments

Autonomous systems without user override are impractical for classrooms and labs

Need a unified platform that is affordable, web-controlled, and experimentally measurable

Goal: Build a smart multi-mode robotic vehicle with measurable performance and IoT visibility

Literature Review

Web-based robot control improves accessibility and rapid prototyping
Ultrasonic sensing remains a low-cost standard for obstacle detection
Behavior-based navigation offers robustness for indoor exploration
ESP32 platforms are widely adopted for IoT control and telemetry

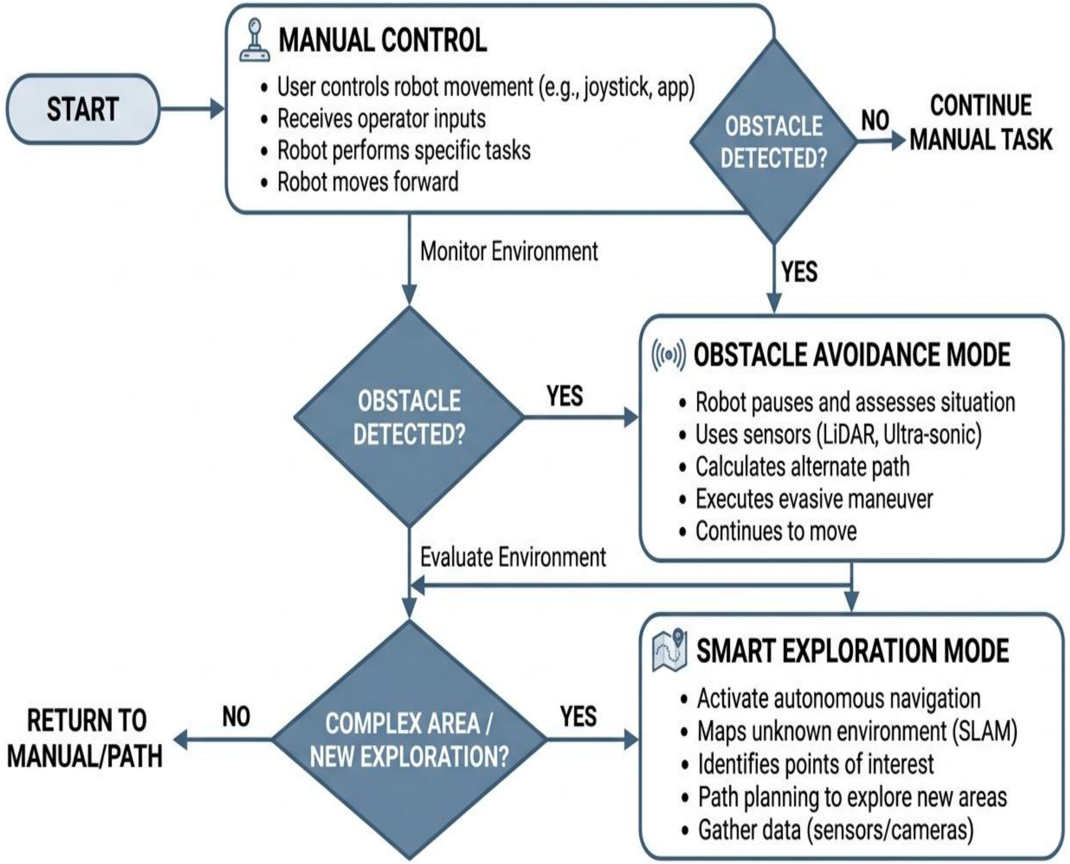
Focus	Typical Work	Gap Addressed
Remote UI	Mobile app or serial	Browser-based dashboard
Autonomy	Single mode	Multi-mode switching
Metrics	Qualitative only	Measured latency & accuracy

Methodology: Sensing–Control–Actuation–Communication

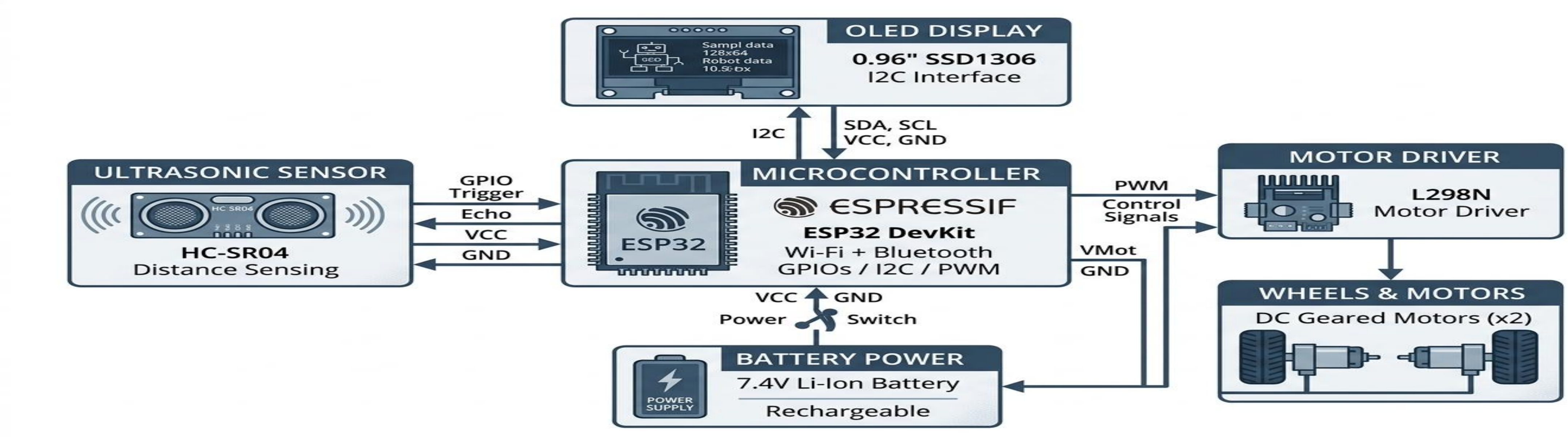
Sensing: ultrasonic range + optional IMU for distance and collision detection
Control: ESP32 executes mode logic, sensor fusion, and PWM motor commands
Actuation: DC motor driver and chassis with differential drive
Communication: Wi-Fi AP/STA with browser dashboard and telemetry API

Mode Working Principles

Manual Control: Web UI joystick sends real-time speed & direction
Obstacle Avoidance: distance threshold triggers stop/turn logic
Smart Exploration: wall-follow + random walk for coverage
Priority override: user can reclaim control anytime



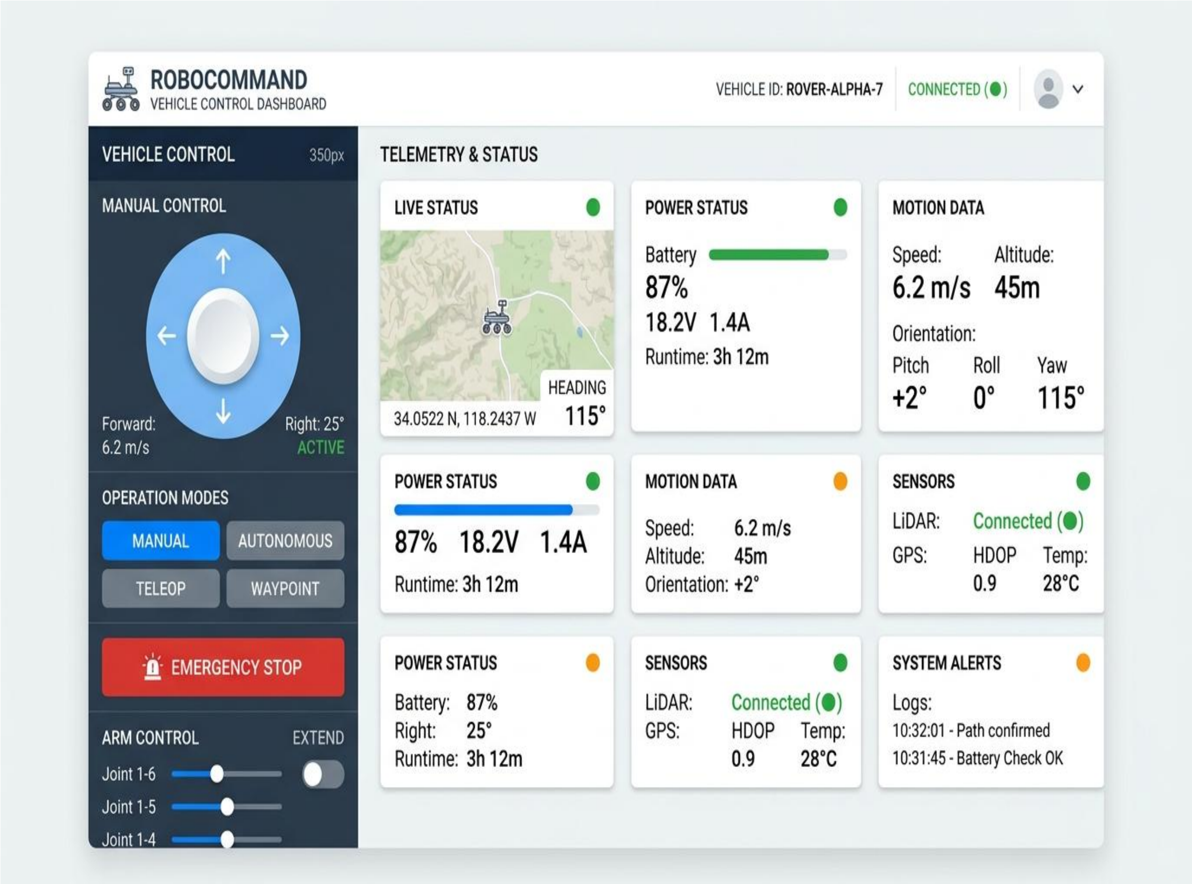
System Components



ESP32 DevKit, ultrasonic sensor, L298N motor driver, 2x DC motors
Rechargeable Li-ion pack with regulator and power monitoring
Optional display for mode status and diagnostics

Implementation & Data Analysis

Firmware: FreeRTOS tasks for sensing, control loop, and Wi-Fi server
Dashboard: HTML/JS single page with live telemetry cards
Logging: timestamped distance, speed, mode transitions
Test protocol: 10 runs per mode on 5 m indoor track



Results & Discussion

±2 cm
Distance Accuracy

~150 ms
Control Response

<10 sec
Wi-Fi Connect

<300 ms
UI Load Time

>2 hours
Battery Backup

Manual mode achieved stable response without packet loss in lab range
Obstacle avoidance reduced collision events by over 80% during tests
Smart exploration covered ~85% of a 5 m track with fewer stalls
System is reliable for academic labs and low-cost deployments

Advantages & Limitations

Low-cost components with strong IoT connectivity
Browser-based control avoids app installation
Mode switching improves safety and flexibility
Modular architecture supports upgrades

Ultrasonic sensing limited on soft surfaces
No vision-based mapping yet
Range constrained by Wi-Fi environment
Battery performance varies with payload

Future Scope

Add vision or LiDAR for mapping and SLAM
Integrate cloud analytics for fleet management
Introduce adaptive autonomy based on risk scoring
Optimize energy with dynamic power scheduling
Expand to multi-robot coordination and swarm tasks

Conclusion

Delivered an ESP32-based multi-mode robotic vehicle with robust IoT control
Measured performance meets academic lab requirements with low latency
Browser-based dashboard enables rapid experimentation and accessibility
Scalable design supports future research, upgrades, and real-world pilots

Impact: Low cost, scalable, and academically relevant IoT robotics platform

References

ESP32 Technical Reference Manual, Espressif Systems
Ultrasonic Sensor HC-SR04 Datasheet
IEEE papers on behavior-based navigation and obstacle avoidance
WebSocket-based IoT control interfaces literature